

Recession Signals Without Recession and the Case of 2024

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Abstract

This paper distinguishes a missed recession from a misprediction. In the latter the indicators err; in the former they perform as designed and the dating authority does not act. The 2022–24 United States cycle is assessed against that standard. A screen of the full 1949–2026 unemployment record establishes that sustained Sahm-rule crossings outside dated recessions are essentially without precedent, 2024 being the sole instance in the real-time series, and three independent instruments corroborate it. A new device, the Bristow Rule, dates a recession’s end at the peak of the Sahm indicator, within three months of every postwar trough: April–August 2024.

On the standard chart of the United States unemployment rate, recessions appear as vertical gray bars marking the peak-to-trough dates chosen by the National Bureau of Economic Research. This paper is about the bars that may be missing. A “missed recession,” as the term is used here, is not a misprediction. A misprediction occurs when indicators point to a downturn that never happens. A missed recession occurs when the indicators fire, economic conditions deteriorate in ways those indicators were built to detect, and the official arbiter nonetheless never dates a contraction. The distinction matters because it locates the fault differently: a misprediction is a failure of the indicator, while a missed recession is a failure of the dating.

From the NBER website, “The NBER’s definition emphasizes that a recession involves a significant decline in economic activity that is spread across the economy and lasts more than a few months. In our interpretation of this definition, we treat the three criteria — depth, diffusion, and duration — as somewhat interchangeable. That is, while each criterion needs to be met individually to some degree, extreme conditions revealed by one criterion may partially offset weaker indications from another. For example, in the case of the February 2020 peak in economic activity, the committee concluded that the subsequent drop in activity had been so great and so widely diffused throughout the economy that, even if it proved to be quite brief, the downturn should be classified as a recession.” The committee adds, on the same page, that “There is no fixed rule about what measures contribute information to the process or how they are weighted in our decisions.”

It matters for a second reason as well. The episodes examined here share a protagonist. In nearly every case, 1966, the recessions the yield curve called correctly, and above all 2022–2024, the sequence begins with the Federal Reserve easing too generously for what later may have been seen as too long, continues with the Fed tightening hard to fight the inflation it had accommodated,

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and ends with the labor market absorbing the damage. When the NBER dates a recession, that damage at least enters the official record and invites scrutiny of the policy that produced it.

A word on dating. The cycle under examination runs from 2022 to 2024; the recession dated inside it is shorter. Every mechanical instrument here places the onset in the first half of 2024 and the end in August 2024, and none flags 2022 or 2023 at all. Throughout, “the post-2022 cycle” means the whole window; “the missed recession” and “the missing gray bar” mean April to August 2024.

The paper makes four contributions. First, it assembles the real-time and current-vintage evidence that the post-2022 cycle contained a recessionary episode. Second, it screens the full 1949–2026 history of the Sahm indicator for threshold crossings outside dated recessions, on both the current vintage and the real-time record, and shows that nearly all are either early warnings of recessions that promptly arrived or the lagging tails of recessions that had just ended; on the real-time record, 2024 is the only sustained standalone crossing there has ever been. Third, it shows that the 2024 signal was not an isolated instrument failure but was seconded by three further mechanical labor-market instruments built by other authors, each with a published record and each firing or coming to its line in the same window. Fourth, it dates both ends of the missing bar with a symmetric pair of mechanical rules — the Sahm Rule for the onset and, for the end, a new rule introduced here as the Bristow Rule, under which a recession ends when the Sahm indicator stops rising — and it draws on the chart, in dotted lines, the gray bar the NBER never did.

2. Data and Methods

All series were retrieved from FRED, the Federal Reserve Bank of St. Louis database, in August 2026: the civilian unemployment rate (UNRATE, monthly, 1948–), the 10-year minus 2-year constant-maturity Treasury spread (T10Y2Y, daily, June 1976–), the 10-year and 1-year constant-maturity yields (GS10, GS1, monthly, used for episodes before the 2-year series begins), real GDP (GDPC1, quarterly, 1947–), real gross domestic income (A261RX1Q020SBEA), real personal income excluding current transfer receipts (W875RX1), real manufacturing and trade industries sales (CMRMTSPL), industrial production (INDPRO), total nonfarm payrolls (PAYEMS), job openings (JTSJOL) and the civilian labor force (CLF16OV), the seasonally adjusted weekly insured unemployment rate (IURSA, 1971–), the effective federal funds rate (FEDFUNDS), the consumer price index (CPIAUCNS), the Federal Reserve’s balance sheet (WALCL), the real-time Sahm indicator (SAHMREALTIME) and its current-vintage counterpart (SAHMCURRENT), and the NBER recession indicator (USREC). Recession dating throughout is the NBER’s. One data note applies: the unemployment rate for October 2025 was never published, because that autumn’s government shutdown suspended the month’s household survey;

where rolling calculations require it, the single missing observation is linearly interpolated, and it is used for nothing else.

The Sahm Rule (Sahm, 2019) signals that a recession has begun when the three-month moving average of the unemployment rate rises 0.50 percentage point or more above the lowest such three-month average of the preceding twelve months. The rule was designed on real-time data, so two versions are tracked: the indicator as originally published (SAHMTIME) and the same statistic recomputed on today’s revised data (SAHMCURRENT).

Three instruments built by other authors are used as independent checks, each computed by its own published rule. The Michez rule (Michaillat and Saez, 2025) takes $\hat{u}$, the three-month unemployment average minus its twelve-month minimum, and $\hat{v}$, the twelve-month maximum of the three-month vacancy average minus its current value, and forms $m = \min(\hat{u}, \hat{v})$; a recession is signaled when m exceeds 0.29 percentage point, with certainty above 0.81 and a probability of $(m - 0.29)/(0.81 - 0.29)$ in between. Because the rule takes the *minimum* of the two channels, a rise in unemployment unaccompanied by a fall in vacancies cannot trigger it — the property that makes it the natural test of the labor-supply explanation of 2024. The Scavette–O’Trakoun–Sahm-style (SOS) indicator applies the Sahm logic to higher-frequency administrative data: it signals when the 26-week moving average of the insured unemployment rate rises by more than 0.20 percentage point above its minimum over the 52 preceding weeks (O’Trakoun and Scavette, 2025). Because insured unemployment counts people actually filing claims after job loss, the SOS is a layoff-based indicator, a distinction that turns out to matter in Section 8. The San Francisco Fed’s Labor Market Stress Indicator (Garimella, Jordà and Singh, 2025) counts the states in which the three-month average unemployment rate stands 0.50 percentage point or more above its prior-twelve-month low, out of fifty-one including the District of Columbia; 30 is its recession line.

For the onset, the Sahm crossing is read as a lagged announcement. Across the eleven recessions from 1953 through 2020 the current-vintage indicator first crossed 0.50 between one and eight months after the NBER’s dated onset, with a median of three and a mean of three and a half; recomputed eleven times, each leaving one recession out, the median is three in every fold. Subtracting that lag dates the 2024 onset to April. On the real-time record the lags run two to four months with a median of four, which dates the onset to March — the month the Michez rule independently produces. For the end, this paper introduces the Bristow Rule: a recession ends when the Sahm indicator stops rising — when its first derivative reaches zero, which for a monthly series means the month of the indicator’s peak, confirmed one release later when the first decline prints. The logic mirrors Sahm’s own construction. Subtracting the trailing twelve-month minimum converts the level of unemployment into a measure of deterioration momentum, and momentum crests when the contraction ends, even where the unemployment rate itself keeps climbing for a year or more afterward.

The rule performs strikingly well: in all twelve postwar recessions, the peak of the current-vintage indicator falls within three months of the NBER trough. It coincides exactly in 1949, 1954, 1980 and 2009; it lags by one to three months in 1958, 1961, 1970, 1975, 1991, 2001 and 2020; and it leads in exactly one case, 1981–82, where the maximum value of 2.50 is first reached in September 1982, two months before the November trough, and then matched again in November, the trough month itself. The median gap is a month and a half; taken without sign, the median is two months.

Three qualifications belong with that record. Three months is not an independently chosen standard: it is the tightest tolerance the record admits, the largest gap being exactly three, in 1990–91. It is nonetheless demanding. Each search window runs between twenty-four and forty months, of which at most seven lie within three months of the trough, so a peak falling at random would meet the tolerance about twenty-three percent of the time, and all twelve would do so by chance with probability of roughly one in sixty million. The rule is specific to troughs rather than to dates in general: scored against false troughs displaced six, twelve, eighteen or twenty-four months in either direction, the same twelve windows return at most one hit.

The record is also a current-vintage record. Run on the real-time series, which reaches nine of the twelve, the rule places the peak within three months in eight; the exception is 1990–91, at four. The confirmation step runs the other way and is stronger on the data of the day: in all nine real-time episodes the month after the peak declines and the peak is never matched again, where on the current vintage it is later matched in three of the twelve. The window definition carries weight as well. It runs from the indicator’s first threshold crossing to twelve months after the crossing episode ends, the definition given in Appendix A.2; anchoring instead on the NBER peak and trough returns the same twelve dates, but widening it to twenty-four months breaks the 1980 case, because the rise belonging to the next recession — whose peak falls twelve months after the 1980 trough — is drawn inside. Finally, the record is postwar. The one out-of-sample test the data allow is mixed: with monthly unemployment from April 1929, the rule lands one month after the 1937–38 trough and eight months before the 1929–33 trough, that contraction having run forty-three months against a twelve-month lookback.

Two properties follow. The record’s single lead belongs to a contraction long enough to outlast the twelve-month lookback: when a recession runs past a year the rolling baseline begins climbing mid-recession and can crest the indicator early, so in every postwar recession shorter than the window the peak arrives at or after the trough and the Bristow date bounds the end from above. The second derivative supplies a herald from the other side: the month the indicator completes its fastest three-month climb lands exactly on the trough in five of the twelve recessions and never runs more than two months past it, but can lead by as much as eleven, so deceleration bounds the end from below rather than fixing it. Together the two rules bracket a recession from both sides

using a single monthly data series, with no committee and no revisions required. For scale, the eleven recessions from 1948 through 2009 lasted between six and eighteen months, with a median of ten; the 2020 contraction, at two months, was a pandemic artifact with no parallel in the record.

3. The Post-2022 Episode

The policy sequence. The episode has an author, and the record of what the author did is not in dispute. Through the end of 2021 the Federal Reserve characterized the fastest inflation in a generation as “transitory.” In prepared testimony before the Senate Banking Committee on November 30, 2021, Powell conceded that “it is difficult to predict the persistence and effects of supply constraints, but it now appears that factors pushing inflation upward will linger well into next year.” Consumer prices were rising 6.8 percent over the preceding twelve months that November, and 7.0 percent by December — the fastest pace since June 1982. Even then the Fed did not tighten. The effective federal funds rate averaged 0.08 percent in February 2022, and the Fed was still expanding its balance sheet, which peaked at \$8.97 trillion in the week ending April 13, 2022 — a month after the first rate hike and with inflation on its way to 9.1 percent that June, the highest since November 1981. Having waited, the Fed then compressed an entire tightening cycle into a year and a half: 5.25 percentage points of target-range increases between March 2022 and August 2023. Measured on the effective rate, the seventeen months to August 2023 delivered a rise of 5.13 percentage points; no seventeen-month stretch had produced as much since the one ending in June 1981. The funds rate then sat at its 5.33 percent peak for more than a year — where it still stood in August 2024, when the Sahm Rule reached its recession threshold. The first cut came only in September 2024. Flood, then slam, then wait: whatever else one concludes about 2022–2024, the timing of the damage examined below is not a mystery.

The yield curve. For half a century, an inverted Treasury curve — short rates above long rates — has been the most watched of all recession signals, and it is at bottom a measure of exactly this kind of policy pressure: the central bank pushes short rates above what the market believes the economy can bear at the long end, credit conditions tighten, and borrowing, investment, hiring, and spending slow. Table 1 lists the record. In the era of the 2-year series — which begins in June 1976 and records no negative day before August 1978 — every inversion episode listed in Table 1 from 1978 through 2019 was followed by a dated recession. The one negative episode of that era not listed, the brief 1998 dip described in the note, was not followed by a recession until 2001. Before 1976, the 10-year–1-year spread inverted ahead of the 1969–70 and 1973–75 recessions as well, and it carries the record’s one sustained false positive — the spread was negative through all twelve months of 1966, and no recession followed — and that exception is taken up in Section 4, because it is the closest historical relative of the present episode.

The 2022 inversion was not marginal. After a two-day dip below zero on April 1 and 4, 2022, the 10Y–2Y spread inverted durably on July 6, 2022, reached its trough of –1.08 percentage points on July 3, 2023, and remained below zero for 537 consecutive trading days through August 26, 2024 — the longest unbroken inversion in the history of the series, exceeding the 423 trading days of 1978–80. The cumulative area below zero over the 2022–24 inversion, at –259 percentage-point days, is the second largest in the series, after the –304 of 1978–80. The curve then flickered around zero and turned durably positive on September 6, 2024. That timing is itself notable: historically, recessions tend to arrive not while the curve is inverted but around the time it re-steepens, which in this cycle was precisely when the labor-market indicators discussed below were crossing their thresholds.

Table 1. Yield-curve inversions and NBER outcomes

Inversion	Outcome	Episode
1966 (10Y–1Y)	No recession	Credit crunch of 1966; growth recession (Sec. 4)
1968–69 (10Y–1Y)	Dec 1969–Nov 1970 recession	1969–70 recession
1973 (10Y–1Y)	Nov 1973–Mar 1975 recession	Oil crisis / stagflation
Aug 1978	Jan 1980–Jul 1980 recession	1980 recession
Sep 1980	Jul 1981–Nov 1982 recession	Volcker disinflation
Dec 1988	Jul 1990–Mar 1991 recession	Early-1990s recession
Feb 2000	Mar 2001–Nov 2001 recession	Dot-com recession
Dec 2005	Dec 2007–Jun 2009 recession	Great Recession
Aug 2019	Feb 2020–Apr 2020 recession	COVID-19 (exogenous shock)
Jul 6, 2022	None dated	Candidate missed recession

Notes: The 2-year constant-maturity series begins in June 1976; earlier episodes are dated with the monthly 10-year–1-year spread (GS10–GS1) and are named for the year or years in which that spread was negative rather than by its first negative month — the 1966 episode, for instance, turned negative in December 1965 and stayed negative through February 1967. From August 1978 onward the entry is the first trading day below zero. The 2s10s spread also closed below zero on 27 trading days between May 26 and July 27, 1998 — in four separate runs, the longest of eighteen days, and never deeper than –0.07 — without a recession following until 2001. The 2019 inversion preceded the pandemic recession, which was triggered by an exogenous shock rather than a credit cycle. Sources: FRED T10Y2Y, GS10, GS1; NBER.

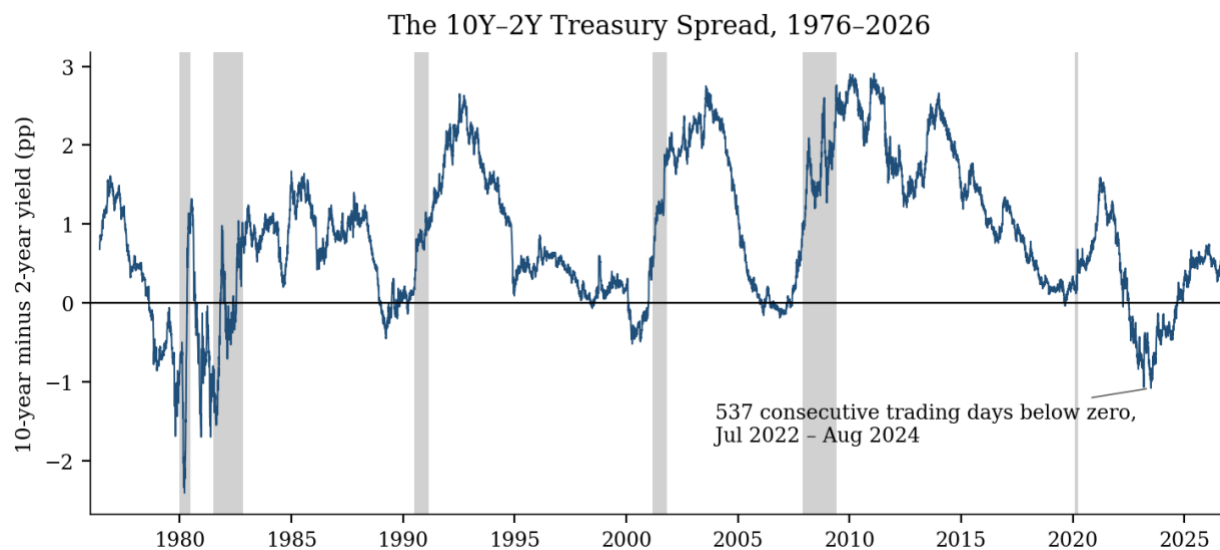


Figure 1. *The 10-year–2-year Treasury spread, 1976–2026, with NBER recessions shaded. The 2022–24 inversion is the longest in the history of the series at 537 consecutive trading days (July 6, 2022 – August 26, 2024), with a trough of -1.08 on July 3, 2023.*

Real-time output. On April 28, 2022, the Bureau of Economic Analysis’s advance estimate reported that “Real gross domestic product (GDP) decreased at an annual rate of 1.4 percent in the first quarter of 2022”; on July 28, 2022, the advance estimate for the second quarter reported a further decrease “at an annual rate of 0.9 percent.” Two consecutive quarters of falling real GDP is the popular “technical recession” definition, and by that definition the United States entered recession in mid-2022. What happened next is the striking part: the contraction was revised away. In the data as they stand in August 2026, the first quarter of 2022 shows -1.0 percent and the second quarter shows $+0.6$ percent. The technical recession of 2022 now exists only in the real-time record. This is not unprecedented — real GDP fell in the second and third quarters of 1947, at annual rates of 1.1 and 0.8 percent in today’s data, and the NBER dated no recession then either — but it underscores the paper’s central point about vintage: households, firms, investors, and policymakers made their 2022 decisions on the data that showed a contraction, not on the revisions that arrived years later.

Nor did the income side of the accounts ever fully recant: real gross domestic income, the conceptual twin of GDP, fell at a 0.3 percent rate in the second quarter of 2022 and a 2.3 percent rate in the fourth, declines that survive in today’s data.

The NBER’s own monthly indicators. The committee does not date recessions from GDP. Its dating-procedure FAQ names six monthly indicators — “real personal income less transfers (PILT), nonfarm payroll employment, real personal consumption expenditures, manufacturing and trade sales adjusted for price changes, employment as measured by the household survey, and industrial production” — while adding that “There is no fixed rule about what measures contribute

information to the process or how they are weighted in our decisions.” Two of those six fell in the first half of 2022 and still fall in the current vintage. Real personal income excluding current transfer receipts declined 1.7 percent between December 2021 and June 2022. Real manufacturing and trade sales fell 2.6 percent from their January 2022 peak to their June 2022 trough. Industrial production peaked in April 2022 and did not regain that level for the rest of the cycle, reaching a trough in January 2024 that stood 2.2 percent below it. This is not a contested reading: it is conceded by the episode’s most careful critic. In his critique of the claim that the United States has been in recession since 2022, Menzie Chinn writes that “manufacturing and trade sales as well as personal income ex.-current transfers dropped in 2022H1,” while noting correctly that both payroll and household employment rose in the same period. He concedes the third of the series examined here in the next sentence: “The only series lower than where it was in mid-2022 is industrial production.”

The labor market. The unemployment rate bottomed at 3.4 percent in April 2023 and drifted upward through 2024. As originally published, the Sahm indicator read 0.53 for July 2024, 0.57 for August, and 0.50 for September — three consecutive months at or above the recession threshold, and the July reading, released on August 2, 2024, was the one that made headlines. That same monthly report showed the unemployment rate at 4.3 percent, with “the number of unemployed people” increasing “by 352,000 to 7.2 million.” Later revisions trimmed the edges — July 2024 now reads 4.2 percent — but the crossing itself survives revision in a form that is, if anything, cleaner: on current-vintage data the indicator reads 0.50 in July 2024, 0.57 in August, and 0.53 in September. Both vintages therefore give three consecutive months at or above the threshold, and both place the peak in the same month, August 2024, at the same value, 0.57. The Sahm Rule fired in the summer of 2024, with the federal funds rate still at its cycle peak, and no recession has been dated. The subsequent path offers no rebuttal: the unemployment rate never returned toward its 2023 lows, reaching 4.5 percent by November 2025 and holding in the low-to-mid 4s through mid-2026.

Labor demand, not merely labor supply. The most common explanation of the 2024 unemployment rise is that it came from a rapidly expanding labor force rather than from weakening demand. The vacancy data do not support reading it as *only* that. Job openings peaked at 12.3 million in March 2022 — a vacancy rate of 7.5 percent of the labor force — and fell to 7.5 million by August 2024, a vacancy rate of 4.5 percent. That is a decline of 39 percent in the level of openings and three full percentage points in the rate. Labor demand was falling steeply at the same time unemployment was rising, which is the configuration of a downturn rather than of a supply-driven measurement artifact. That the two would move together was the prediction rather than the surprise. Writing in July 2022, Blanchard, Domash and Summers argued that “fighting inflation will require a reduction in job vacancies and also an increase in unemployment,” and that “the Fed’s hope that vacancies can be decreased without increasing unemployment flies in the face of

historical empirical evidence.” The vacancy rate then fell three points, to 4.0 percent by November 2025, while unemployment climbed from 3.4 percent in April 2023 to 4.2 by August 2024 and 4.5 by November 2025: the joint movement they called unavoidable is the one the data record. Section 7 shows that the published instrument designed precisely to separate the two cases reaches the same conclusion.

What the payroll data later revealed. Revisions cut both ways in this cycle, and in the labor data they cut against the economy. The Bureau of Labor Statistics’ annual benchmark, which replaces survey estimates with the near-universe of unemployment-insurance tax records, revised the level of nonfarm employment for March 2024 down by 598,000 on a not-seasonally-adjusted basis, published February 7, 2025, and the level for March 2025 down by 862,000 not seasonally adjusted — 0.5 percent, in the Bureau’s own statement of it; the PDF edition and table 4 of the same article report 861,000, after accounting for a reconstruction affecting two banking industries — and 898,000 seasonally adjusted, published February 11, 2026. The labor market of 2023–2025 was materially weaker than the data available at the time showed. The same vintage problem that erased the 2022 output contraction after the fact concealed part of the 2024 employment shortfall while it was happening. Table 2 sets the episode’s real-time and current-vintage readings side by side.

Table 2. The post-2022 evidence: real time versus current vintage

Indicator	Real-time reading	Current vintage (August 2026)
10Y–2Y spread	Inverted Jul 6, 2022; trough –1.08 (Jul 3, 2023)	Unrevised; 537 trading days below zero — longest in series history
Real GDP, 2022 H1	Q1 –1.4% (advance); Q2 –0.9%	Q1 –1.0%; Q2 +0.6% — contraction revised away
Real GDI, 2022	—	Q2 –0.3%; Q4 –2.3% — declines survive
Real personal income less transfers	—	–1.7% Dec 2021 → Jun 2022
Real manufacturing and trade sales	—	–2.6% Jan 2022 → Jun 2022
Industrial production	—	Peak Apr 2022; –2.2% to Jan 2024 trough
Job openings	12.3 mn (Mar 2022)	7.5 mn (Aug 2024); vacancy rate 7.5% → 4.5%
Sahm Rule, 2024	0.53 (Jul), 0.57 (Aug), 0.50 (Sep)	0.50 (Jul), 0.57 (Aug), 0.53 (Sep)
Michez rule, 2024	—	Reaches 0.29 — detection — in Mar 2024; peaks 0.54 in Aug 2024
Unemployment, Jul 2024	4.3%; unemployed +352,000 to 7.2 million	Revised to 4.2%; 4.5% by Nov 2025
Nonfarm payrolls, Mar 2024 / Mar 2025	—	Benchmarked down 598,000 / 862,000 (NSA)

SOS indicator, 2023	—	Plateaus at exactly 0.2000 for twelve weeks, Sep 2–Nov 18, 2023; never exceeds it
NBER dating	—	No recession dated after Feb–Apr 2020

Sources: FRED (T10Y2Y, GDPC1, A261RX1Q020SBEA, W875RX1, CMRMTSPL, INDPRO, JTSJOL, CLF16OV, UNRATE, SAHMOREALTIME, SAHMCURRENT, IURSA); BEA advance estimates of April 28 and July 28, 2022; BLS Employment Situation, July 2024 (released August 2, 2024) and CES benchmark articles of February 7, 2025 and February 11, 2026; Michaillat and Saez business-cycle dashboard; NBER; authors’ calculations.

4. Are There Other Missing Gray Bars?

If the Sahm Rule is the dating instrument, the honest procedure is to apply it everywhere and see what it flags. The screen therefore covers every month from 1949 through July 2026 in which the current-vintage indicator sits at or above 0.50 while the NBER dates no recession. Contiguous months are grouped into episodes, and each episode is sorted by a mechanical criterion rather than by judgment: an episode that begins within a month of an NBER trough is a lagging tail; one that ends within five months of an NBER peak is an early warning; anything else is standalone. Neither window drives the result: widening the tail window to three months or the warning window to seven leaves the same two standalone episodes, and tightening the warning window to three months adds only the November–December 1959 pair, which was followed by a recession five months later. The screen returns three kinds of episodes.

The first is a single early warning: the indicator crossed in November and December of 1959, at 0.60 and 0.53, five months before the recession of April 1960 began — a period distorted by the basic steel strike, which the Bureau of Labor Statistics records as having “idled 519,000 workers for 116 days” from July 15, 1959 — and a signal that was early rather than wrong.

The second, and by far the largest, group is the lagging tails: after every recession from 1949 through 2020, the indicator remains elevated for months because unemployment peaks well after output troughs. Twelve such tails appear, one for each postwar recession. The long 1991–92 stretch — the jobless recovery of that cycle, in which the indicator stayed above threshold through October 1992, nineteen months after the recession formally ended — belongs to this group, as does the 2001–02 aftermath, which ran through November 2002.

That leaves exactly two standalone crossings in seventy-seven years. The first is July–August 2003, when the indicator sat at precisely 0.50 for two months, twenty months after the 2001 recession ended and with the unemployment rate peaking at 6.3 percent in June 2003 (up from 5.5 percent at the recession’s November 2001 trough). The second is July–September 2024. The two are not alike. The 2003 crossing was marginal — exactly at threshold, never above it — and everything around it argues for reading it as the last shadow of the 2001 downturn rather than a fresh one: the unemployment rate began falling immediately afterward, real GDP grew at a 6.8

percent annual rate in the third quarter of 2003, and the yield curve was steep. The 2024 crossing cleared the threshold on both vintages, ran for three months rather than two, arrived on the heels of the longest curve inversion on record and a real-time output contraction, was seconded by three independent labor-market instruments, and was followed not by a boom but by an unemployment rate that kept drifting up.

The screen is sharper still when run on the real-time record — the readings as they were actually published, which is what a policymaker or forecaster would have seen. The SAHMOREALTIME series begins in December 1959. Outside dated recessions and their lagging tails, it produces exactly four episodes in sixty-seven years. Two are early warnings: December 1959 and January 1960, at 0.77 and 0.50, four months before the April 1960 recession began; and October 1969, a single month at 0.50, two months before the December 1969 recession began. One is a single month at exactly 0.50 in November 1976, of which Sahm herself writes, “November 1976 at 0.50 (which, when unrounded, does not trigger).” The fourth is July through September 2024, three consecutive months with a peak of 0.57 — the only one of the four that is neither an early warning nor an isolated single month. The 2003 episode does not appear at all in the real-time record — as published, the indicator peaked at 0.47 that July and August and never crossed. On the record as it was actually published, the 2024 episode is the only crossing of more than one month that is neither the tail of a recession just ended nor the herald of one about to begin.

A Federal Reserve economist has audited the same record and read two of these episodes differently. Atkinson (2024), writing for the Dallas Fed two months after the 2024 crossing, marks “episodes when the threshold was crossed (November 1959 and November 1976) or nearly crossed (October 1967 and July 2003) without a recession or further sharp increase in unemployment.” On the first the data do not support that reading: the NBER dates a recession from April 1960, five months after the November 1959 crossing, which is why the screen here files it as an early warning rather than a false alarm. On the second the rule’s author is the authority against it, since Sahm writes of November 1976 “at 0.50 (which, when unrounded, does not trigger).” What Atkinson establishes, and what this paper adopts rather than disputes, is the timing property: “The fact that the signal is contemporaneous rather than leading reflects another aspect of economic forecasting: Recessions are very difficult to predict.” A coincident signal is exactly what the dating convention of Section 2 assumes, and it is why a crossing is read here as the announcement of a downturn already some months old. His grounds for expecting no recession in 2024 — “ample GDP growth, stable layoffs and growing real household wealth” — are the three this paper takes up in Sections 8 and 9: output as published against output as revised, a downturn transmitted through hiring rather than layoffs, and a household sector cushioned by transfers the Federal Reserve did not make.

The screen is equally instructive for what it excludes. Three well-known soft patches show sustained or partial upturns in unemployment without ever approaching the Sahm threshold, and

under the rule they earn no bar. In 1966–67, the rate rose from 3.6 percent in November 1966 to 4.0 percent in October 1967, a Sahm peak of just 0.23 — yet this is the one episode in the record where the yield curve also inverted, for all twelve months of 1966, with no recession following. The script should by now be familiar. Burger’s contemporaneous account names the two considerations that dominated stabilization policy that year — “(1) the sharply rising level of Government spending for the Vietnam war, and (2) the emergence of inflation” — and records that Federal Reserve actions “became much more restrictive in July through August 1966, the period of the so-called ‘Credit Crunch,’” in which “the growth of bank credit came to a temporary halt” (Burger, 1969). A fiscal expansion accommodated, then tightened against; and no recession dated. It is the classic “growth recession,” the closest historical analogue to 2022–24, and an early instance of the pattern this paper documents: a hard tightening that dented the economy while the official record recorded nothing.

In 1985–86, the unemployment rate stopped improving amid the oil-price collapse and farm-belt stress, dipping to 6.7 percent in January 1986, bouncing to 7.2, and flattening near 7 for most of the year, a Sahm peak of 0.27 — a stall, not a downturn. And in 1962–63 the rate edged from 5.4 to 5.9 percent, a Sahm peak of 0.30. Modest wobbles of this kind are common; standalone threshold crossings are not. The rarity can be counted rather than asserted. Of the 808 months since March 1949 that fall outside a dated recession, 129 sit at or above 0.50 — but 122 of those are the lagging tails of recessions just ended and two are early warnings of recessions about to begin, leaving five standalone months, or 0.62 percent of the non-recession record, about one month in a hundred and sixty. On the real-time series the figure is four months of 704, 0.57 percent, and only one of the two standalone episodes there lasts longer than a single month. That rarity is what makes 2024 remarkable.

The rarity can be put in probability terms. Hornstein (2023) calibrates the false-positive rate of unemployment-based recession signals directly: a year-over-year rise of 0.20 percentage point in the unemployment rate corresponds to a 10 percent false-positive rate, 0.40 points to 5 percent, and 1.30 points to 1 percent, with an area under the ROC curve of 0.94 for the unemployment rate alone and 0.99 when a six-month-lagged 10-year–3-month spread is added. The 2023–24 rise of roughly a full percentage point from the April 2023 low sits far into the tail of that distribution — and it arrived alongside the very curve inversion that Hornstein finds raises the discriminatory power of the unemployment signal to 0.99.

5. Drawing the Missing Gray Bars

Figure 2 puts the two rules to work. The dotted box on the right is the missing gray bar of the current cycle. Its onset follows from the Sahm Rule: the first threshold crossing occurs in July 2024 on both vintages, and subtracting the median three-month lag between the NBER’s dated

onset and the indicator's first crossing dates the start to April 2024. The independent instrument agrees to within a month. Michaillat and Saez date a start by their own convention — “The detected recession start date is the month when the indicator crosses the 0.29pp threshold from below” — and on their dashboard that month is March 2024. Two mechanical rules, built on different data by different authors, therefore place the onset in March and April 2024 respectively, and place the peak in the same month, August. Its end follows from the Bristow Rule, and here the two data vintages — which disagree about nearly everything else in this episode — agree exactly: the indicator peaked in August 2024 on both, at 0.57 on the data of the day and 0.57 on today's data, declined the following month on both, fell back below the threshold in October 2024 on both and never again approached it: on today's data 0.40 is its highest subsequent reading, easing to 0.13 by August 2025 before a modest late-2025 rise to 0.35 and falling to -0.03 by July 2026; on the data as published the corresponding readings are 0.43, 0.13, 0.43 and -0.03 . The first derivative reached zero in August 2024 on both vintages, and the bar closes there: April to August 2024, four months peak to trough, briefer than any postwar recession except the pandemic's two months, and the mildest on record, with a peak indicator reading barely more than a third of the smallest peak in any dated postwar recession (1.50, in 1990–91).

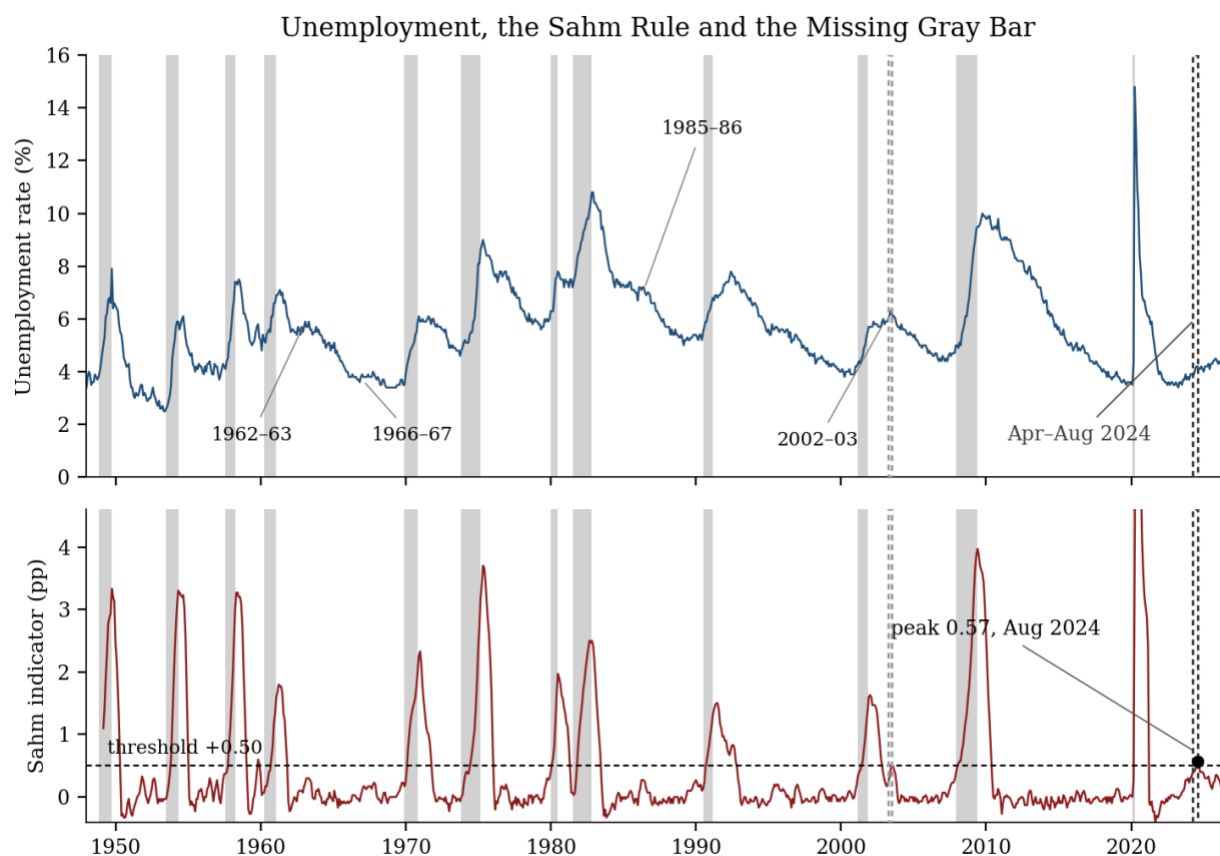


Figure 2. Top: the unemployment rate, 1948–2026, with NBER recessions shaded; the three sub-threshold soft patches discussed in Section 4 — 1962–63, 1966–67 and 1985–86 — are annotated, as is the 2002–03 jobless

recovery. The darker dotted box is the hypothesized missed recession, April to August 2024: onset dated by the Sahm Rule, end by the Bristow Rule — the month the Sahm indicator peaked on both data vintages. The lighter dotted box is the marginal 2003 crossing, April to July 2003 under the same two rules, drawn for completeness and discounted in the text. Bottom: the current-vintage Sahm indicator with its +0.50 threshold; the marked point is the August 2024 peak, where the indicator's first derivative reaches zero. The October 2025 unemployment observation was never published (government shutdown) and is interpolated for the rolling computation only.

The brevity and the mildness are not embarrassments to the dating; they are the explanation for the miss. Nor is the subsequent labor market a rebuttal: unemployment is a lagging series — it kept rising for fifteen months after the 1990–91 trough and for nineteen after 2001 — so the drift to 4.5 percent by November 2025 is the familiar echo of a contraction already ended, not evidence of one still underway. The second derivative agrees: over the rule's own window — the crossing episode of July to September 2024 and the twelve months after it — the indicator's fastest three-month climb ends in August 2024, the same month the level peaks. There is an irony in the timing: the Sahm alarm rang, with the funds rate still at its cycle peak, in the very month the Bristow Rule says the episode was ending. For completeness, a second, lighter box marks the marginal 2003 crossing under the same two rules (onset April 2003, end July 2003 — the indicator reads 0.50 in July and August alike, and the rule resolves the tie to the first occurrence); the rules flag it and the text above discounts it. The 1966–67, 1985–86 and 1962–63 episodes are annotated but receive no box, because the rules do not flag them.

The screen is as important for the years it does not flag as for the one it does, and the two years that a reader might expect to see dated are precisely the two the rules refuse. In 2022 — the year of the two negative quarters of real-time GDP, the falling NBER monthly indicators and the start of the inversion — the Sahm indicator never rose above 0.03 and the Mischez indicator never above 0.04. Nothing remotely resembling a labor-market contraction is present, and payroll employment rose 1.7 percent in the first half of that year. In 2023, the year the unemployment rate began climbing from its April trough of 3.4 percent, the Sahm indicator peaked at 0.30 and the Mischez indicator at 0.26, both below their thresholds, and the SOS indicator touched its line without crossing it. Neither year is dated here, and neither can be dated by these rules. What 2022 supplies is the cause and the real-time output evidence; what 2023 supplies is the approach. The contraction itself is a 2024 event, and the instruments say so with one voice.

6. The SOS Indicator: A Near-Signal, and Why It Matters

The SOS indicator, developed by John O'Trakoun of the Richmond Fed and Adam Scavette of the Philadelphia Fed, translates the Sahm logic to weekly unemployment-insurance data and, per the Richmond Fed, “identifies the past seven U.S. recessions going back to the early 1970s” (O'Trakoun and Scavette, 2025). Recomputed on current-vintage data, the indicator rose steadily through 2023 and then did something almost theatrical: it plateaued at exactly 0.20 — the threshold itself, to four decimal places — for the twelve weeks ending September 2 through November 18,

2023, touching the line without ever rising above it (Figure 3). Under the indicator's strict definition, which requires a rise of *more* than 0.20, the SOS never signaled. It is reported here as a near-signal rather than a signal, a correction of an earlier draft that described the threshold as crossed rather than touched.

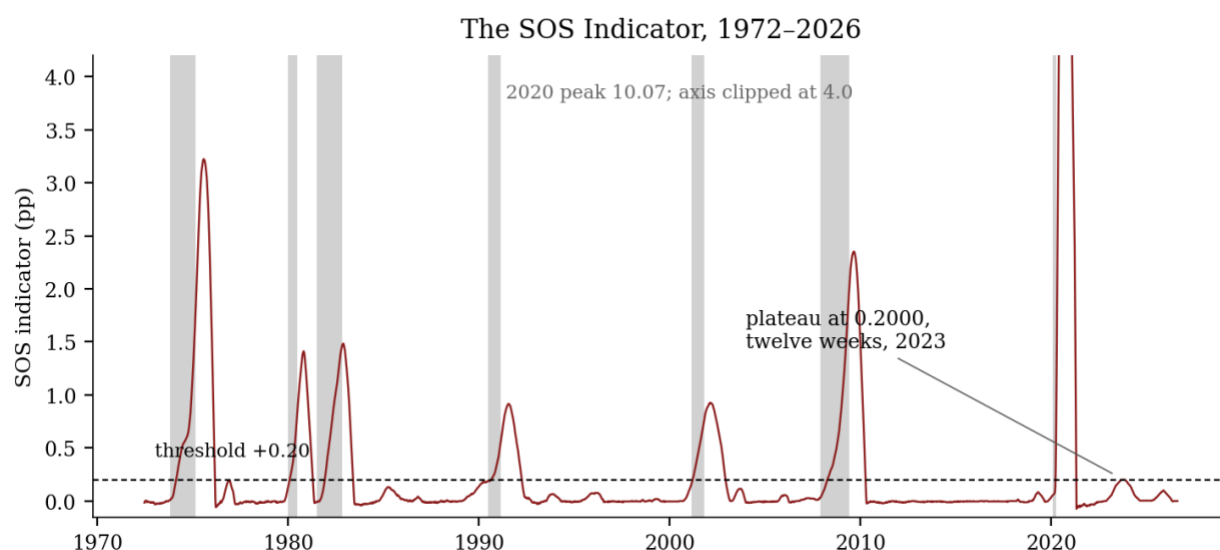


Figure 3. *The SOS indicator, 1972–2026, computed from the weekly insured unemployment rate (26-week moving average minus its 52-week minimum), with NBER recessions shaded. The indicator exceeded its +0.20 threshold in each of the seven dated recessions in the sample; in 2023 it plateaued at exactly 0.2000 for twelve weeks, from September 2 through November 18, without exceeding it, and stood at 0.0 as of the week ending August 15, 2026. The vertical axis is clipped at 4.0 so the pre-pandemic record stays legible; the 2020 reading reaches 10.07.*

The near-miss is diagnostic, not merely a curiosity. The Sahm Rule, built on the household survey, counts everyone unemployed regardless of why; the SOS, built on insurance claims, effectively counts layoffs. In 2023–24 the first crossed and the second did not, which says the rise in unemployment came disproportionately from slower hiring, new entrants, and a rapidly expanding labor force rather than from job destruction. Firms that had just come through a labor market in which vacancies peaked at 7.5 percent of the labor force hoarded the workers they had; layoffs, and therefore insured unemployment, stayed near record lows even as the number of unemployed people grew. A downturn of that shape mutes every layoff-based alarm the profession has, and it goes some way toward explaining how a recessionary episode could slip past the official dating.

7. Corroboration: Independent Instruments in the Same Window

The 2024 signal was not one indicator misbehaving. Three further mechanical instruments, built by other authors, each with its own data, its own rule, and its own published record, read the episode as recessionary; two of the three fired or came to their line in the dated window itself, and

the third, whose published readings fall in 2025, is discussed with that difference stated. Two further published results bear on the same episode and are taken up after them: a postwar regularity that the current cycle breaks unless a recession occurred, and a leading composite that had issued a formal recession call for the quarters immediately preceding.

The Michez rule. Michaillat and Saez (2025), published in the *Oxford Bulletin of Economics and Statistics*, introduce a refinement of the Sahm Rule that takes the minimum of an unemployment indicator and a vacancy indicator. The construction is deliberately conservative: because the statistic is a minimum, a rise in unemployment that is *not* accompanied by a fall in vacancies cannot trigger it. The authors report that the rule “perfectly detects the 15 recessions that occurred between 1929 and 2021, without producing any false positives or negatives,” and that “it detects the nine recessions that occurred between 1960 and 2021 with an average delay of 1.2 months, while the Sahm rule detects them with an average delay of 2.7 months.” Their conclusion on this cycle is stated plainly: “The Michez rule suggests that the US economy entered a recession in 2024.”

On the authors’ own continuously maintained dashboard, read in August 2026, the indicator reaches exactly 0.29 — the threshold — in March 2024, rises above it in April, and stays at or above it for fifteen consecutive months, through May 2025, peaking at 0.54 in August 2024, an implied recession probability of 48 percent under their dual-threshold extension. It falls back below the line in June 2025, returns to it briefly at 0.31 in December 2025 and January 2026 — the counterpart of the modest late-2025 rise in the Sahm indicator noted in Section 5 — and reads 0.00 in June and July 2026. The paper itself states that “The Michez rule first detected a recession in March 2024, as the minimum indicator reached 0.29pp then.” An independent recomputation of the rule from FRED primary data (JTSJOL, CLF16OV, UNRATE) reproduces the same shape — a first month above the threshold in December 2023, a one-month lapse in January, a sustained crossing from February 2024, a peak in August 2024, and a return below threshold during 2025 — though not the authors’ exact values, for two documented reasons: the dashboard assigns each month the job openings the BLS assigns to the previous month, to align the JOLTS reference day with the CPS reference week, and it computes the unemployment rate at basis-point precision from the unemployed count and the labor force rather than from the published one-decimal rate. Applying those two conventions to the same FRED series reproduces the dashboard indicator in every month from February 2002 to September 2025 but one; in 2024 the difference is entirely in the unemployment channel, and the vacancy channel never binds on either construction.

Three features of this deserve emphasis. First, an instrument with a perfect in-sample record over fifteen recessions and ninety-two years signaled a recession in 2024. Second, it peaked in August 2024 — the same month the Sahm indicator peaked, and the month the Bristow Rule dates as the end of the episode. Two independent labor-market statistics, one built on the household

survey alone and one combining it with the firm-side vacancy survey, crest in the same month. Third, and most importantly for the interpretive question, the vacancy channel never bound. Recomputed from FRED by the formula set out in Appendix A.2, the vacancy indicator never fell below 0.71 percentage point in any month of 2024 — more than twice the 0.29 threshold, and 0.94 or above in every month from February through October — so in all twelve months the minimum was set by unemployment. The gate designed to refuse a labor-supply-driven signal did not refuse this one, because vacancies were in fact collapsing.

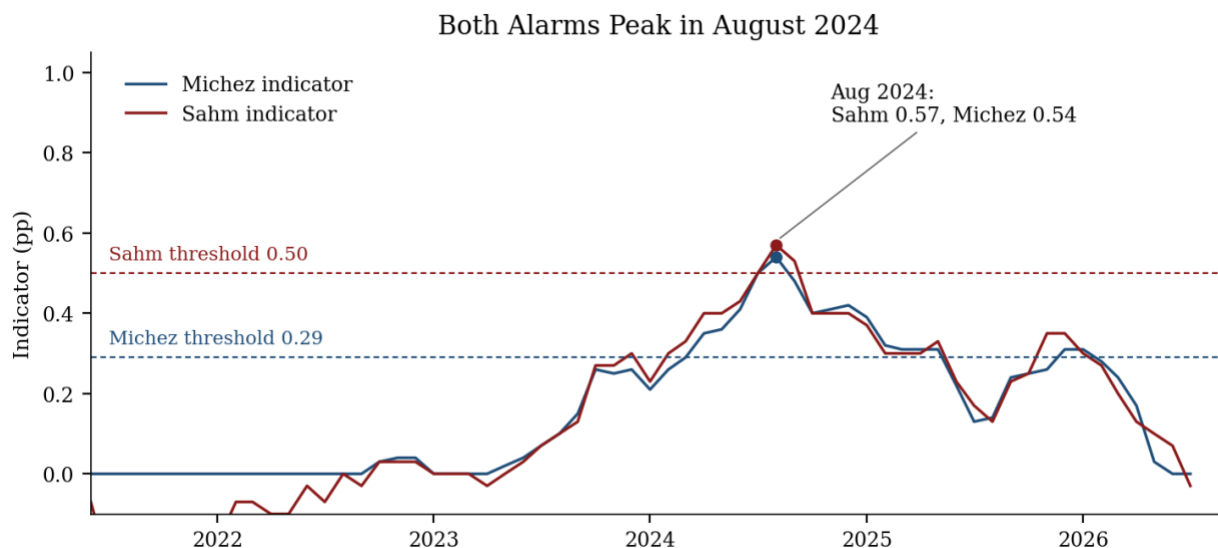


Figure 4. *The Michetz indicator — the minimum of an unemployment indicator and a vacancy indicator (Michaillat and Saez, 2025), read from the authors’ Automated Business Cycle Dashboard in August 2026 — plotted against the current-vintage Sahm indicator, with each rule’s own threshold. The Michetz rule reaches 0.29 in March 2024 and stays at or above it for fifteen consecutive months, through May 2025. Both indicators peak in August 2024, at 0.54 and 0.57 respectively.*

The classifier ensemble. Michaillat (2025) constructs “millions of recession classifiers” from unemployment and vacancy data — hundreds of millions in the later arXiv version — keeps only those identifying “all 15 historical recessions in the 1929–2021 training period without any false positives,” and takes the vote of the high-precision frontier. The NBER version reports that the ensemble “signals recessions 2.2 months after their true onset, with a standard deviation of detection errors of 1.9 months,” and that “applied to May 2025 data, the classifier ensemble gives a 71% probability that the US economy is currently in recession”; the arXiv version gives 2.1 months, a 1.8-month standard deviation, and a 64 percent probability on September 2025 data. The same version reports that “between 1979 and 2021, the committee takes on average 6.3 months to determine recession starts, while the classifier ensemble only takes 1.2 months.” Two cautions travel with the reading. The backtests are sensitive to the training window: the NBER version reports current-recession probabilities of 58, 83 and 25 percent on 1929–2004, 1929–1984 and 1929–1964 training data, while the arXiv version’s backtests, run on five longer windows, give “a

current recession probability between 50% and 67%.” And the readings themselves point to a longer or later episode than the one dated here.

That divergence should be stated rather than smoothed over. What they establish for present purposes is narrower and still substantial: the mechanical labor-market evidence over 2024–2025, read by a method with no in-sample false positives in ninety-two years, was recessionary, and it was not close.

The thirty-state rule. Garimella, Jordà and Singh (2025), writing for the Federal Reserve Bank of San Francisco, count the states in which the three-month average unemployment rate stands at least half a point above its prior-twelve-month low. Their claim is unambiguous: “The national economy has invariably been in recession each time 30 or more states simultaneously experienced accelerating unemployment.” Their companion working paper sharpens it: “the indicator never crosses 28 during non-recessionary periods,” and “the highest non-recession peak occurred in November 1959 with 27 states.”

In 2024 the indicator came to that line. The *Economic Letter* of August 18, 2025 reports that in the July 2024 episode “the share of states meeting our LMSI-based rule barely cleared the 30-state threshold and quickly fell within one month,” and that “nearly 70% of the labor force resided in a state with accelerating unemployment, slightly below the 75% mark typically associated with recession calls.” The working paper of December 2025, computed on later-vintage data, instead describes “the July 2024 episode, when 25 states and about 47 percent of the labor force were in the accelerating bin,” and states that in that month “the indicator came close to the lower threshold.” The two documents are by the same authors and they do not agree; the discrepancy is itself an instance of this paper’s theme, since on one vintage the rule barely cleared its line and on another it stopped five states short of it. Both prose accounts name July; the authors’ own published monthly series names August, reading 17 states in July 2024 and 30 in August, the only month since March 2021 to reach the line. On either reading, a breadth instrument whose authors say it has never been crossed outside a recession came at or across its line in the same month the Sahm and Mischez indicators peaked.

The low-base regularity, and its break. Macaluso (2026), writing for the Richmond Fed, states the historical regularity plainly: “Historically, every prior episode of unemployment rising from a low base while still under 4.5 percent was followed by a recession.” The current episode ran from a 3.4 percent trough in April 2023 to 4.4 percent in December 2025 — 29 months below 4.5 percent, which she describes as “the longest such episode in the post-World War II era,” and 33 months of rising unemployment in all. Either the regularity has broken for the first time since the Second World War, or a recession occurred and was not dated. This paper takes the second reading; the point here is only that a Federal Reserve Bank has published the choice.

The Conference Board’s leading index. The Conference Board’s Leading Economic Index issued a formal recession call during the same cycle. Its release of July 20, 2023 reported that “The Leading Index has been in decline for fifteen months — the longest streak of consecutive decreases since 2007-08, during the runup to the Great Recession,” and stated: “We forecast that the US economy is likely to be in recession from Q3 2023 to Q1 2024.” The index withdrew that call during 2024, a reversal recorded in Section 9.

8. Why the Recession Was Missed — and Why No One Answered for It

Five features of the episode, taken together, account for the missing bar.

First, revisions — in both directions. The two negative quarters that defined the 2022 technical recession were later revised to -1.0 and $+0.6$ percent; the NBER, which dates cycles retrospectively on revised data, was looking at a different 2022 than the one contemporaries lived through. The income side kept part of the story — real GDI still shows declines in the second and fourth quarters of 2022 — and the resulting GDP–GDI divergence is itself a symptom of an economy hard to measure in real time. Revisions cut the other way in the labor data: the annual benchmarks removed 598,000 jobs from the March 2024 level and 862,000 from March 2025, so the employment strength that argued against a downturn was partly an artifact of the vintage available at the time.

Second, the contraction rolled through sectors rather than striking them at once. Housing buckled first under the steepest policy tightening since 1981: by August 2022 the National Association of Home Builders reported that builder confidence “fell for the eighth straight month in August, falling six points to 49 and marking the first time since May 2020 the index fell below the key break-even measure of 50,” with single-family starts down 10.1 percent to a 916,000 annual rate and new and existing home sales down 29.6 and 20.2 percent from a year earlier — a configuration the association itself called an “ongoing housing recession.” Freight followed: the Cass Transportation Index Report of February 2024 noted that “It’s been over two years since the first y/y decline of this freight recession,” with shipments down 5.5 percent in 2023 after rising 0.6 percent in 2022. Manufacturing followed too: the Institute for Supply Management’s Manufacturing PMI (ISM) fell below 50 in November 2022 “for the first time since May 2020 after 29 consecutive months of growth,” and was still reporting its “eighth consecutive month” of contraction in November 2024. Meanwhile services, still reopening, expanded. The NBER’s working definition requires a significant decline in activity that is spread across the economy; a sequence of sectoral recessions, each offset by a booming sector elsewhere, can fail the diffusion test at every point in time even if most of the economy contracts at some point in the window.

Third, the labor market deteriorated in an unusual way. Unemployment rose roughly a point from its 2023 low, but substantially through labor-force growth — Sahm points to “the jump in

immigration in recent years” — and lengthening job search rather than through layoffs. That is precisely the configuration in which the Sahm Rule fires and the SOS does not, and it is a configuration that softens the “depth” the NBER looks for in employment-based measures.

Fourth, the household sector was unusually insulated: pandemic-era fiscal transfers and the stock of excess savings cushioned consumption through the tightening, so the demand collapse that ordinarily converts a slowdown into an unmistakable recession never fully arrived.

Fifth, the deterioration was concentrated in labor demand, where the official chronology is not built to look. Brave, Crust, Eusepi, Hobijn and Şahin (2026), in the *Brookings Papers on Economic Activity*, decompose ninety-four labor-market indicators and find that “Beginning in mid-2022, labor demand started to decline which in magnitude was similar to recessions but unfolding more gradually.” Their interpretation is the opposite of this paper’s — they read “recession-like declines in labor demand without a sharp increase in unemployment rate” as “a distinguishing feature of a soft landing relative to a recession” — but the descriptive finding is not in dispute. What is in dispute is whether an episode in which labor demand falls at recession-like rates, output contracts in real time, the curve inverts for longer than ever recorded, and four labor-market alarms fire is best called a soft landing because the unemployment rate rose only a point.

It is worth being direct about who benefits. None of the cushions just listed was of the Federal Reserve’s making: the fiscal transfers came from Congress, the labor hoarding from firms scarred by the reopening, the labor-supply surge from immigration. The Fed supplied the shock; the rescue came from elsewhere. Yet because the arbiter never drew the bar, the episode has entered the narrative as a “soft landing” — a term that awards the central bank credit for the resilience of an economy it nearly broke. The term does less work than it appears to. Blinder (2023), in the *Journal of Economic Perspectives*, studies “the eleven episodes of monetary policy tightening by the Fed since 1965” and argues that the Fed’s record is better than the conventional view that soft landings are “difficult to achieve and extremely rare” — but he attaches the qualification that matters here, that this holds “as long as the criteria for softness are not too stringent,” and that a soft landing requires “both skill in managing monetary policy and the absence of adverse external shocks.” Softness, on the most sympathetic published account of the Fed’s record, is a graded judgment rather than a fact read off the data, and in practice the criterion that settles it is whether a gray bar was drawn. The asymmetry is structural: recessions are dated years later, on revised data, by a committee with no mandate to assign cause, while central bankers narrate their own performance in real time. An institution whose errors are recorded only when a gray bar appears has every incentive to prefer a chart without one. The claim is not that the dating was manipulated — the revisions, the diffusion problem, and the odd shape of the labor-market damage are genuine — only that the current framework reliably converts near-misses into exonerations.

9. Objections, and What the Data Say

The case above is contested, and the objections deserve to be met on their own terms rather than paraphrased away.

“Claudia Sahm says it is not a recession.” She does, and her reasoning is specific: in July 2024 she wrote that shifts in the labor force were “likely affecting the change in the unemployment rate in a way that would not be typical in prior business cycles,” and that the entrant share was unchanged, “consistent with increased labor supply from immigrants pushing up unemployment and not a sign of weakening demand.” The author of a rule is entitled to say when she thinks it is misfiring. Three things cut against the labor-supply reading. Vacancies: job openings fell 39 percent between March 2022 and August 2024, and the vacancy rate fell from 7.5 to 4.5 percent — falling demand, not merely rising supply. The Michez rule exists precisely to make that distinction, cannot fire on rising unemployment alone, and fired anyway. And the cross-state evidence: de Soyres, Fleck, Peterman, Pradhan and Ratner (2026), applying a state-level Sahm rule at the Federal Reserve Board, report that across the fastest- and slowest-growing population quintiles, pooled over the national recessions of 1980 to 2007, “Despite experiencing starkly different population, labor force, and employment outcomes, the unemployment rate dynamics are remarkably similar, both before and during the recession.”

“The Budget Lab scores 2024 as a false positive.” Demianczuk, Gupta and Nunn (2026) write that the labor market’s normalization “eventually triggered the Sahm Rule in July 2024,” but that “in 2024 and 2025, no recession ensued,” and that for the Michez rule “its recent false positive was particularly severe... the Michez Rule fired in March 2024 and remained above its threshold into 2025.” Two observations. First, this scoring assumes the conclusion: an indicator is a false positive only if no recession occurred, which is the question at issue. Second, the same piece concedes the rarity that makes the episode interesting, noting that a Sahm firing has “virtually never occurred in the post-war period without a recession ensuing.” That is the paper’s point stated by an opponent.

“The San Francisco Fed calls its own 2024 signal a false alarm.” Its stated grounds are that the signal was brief and geographically limited: the count “quickly fell within one month,” and labor-force coverage reached 70 percent rather than the 75 percent “typically associated with recession calls.” But brevity is precisely what this paper argues characterizes the episode — a four-month contraction, the mildest on record. Rejecting a signal because it did not last long is not an independent test when the hypothesis is that the downturn was short. And the Bank’s own published monthly series places the crossing in August 2024 — 17 states in July, 30 in August, the only month since March 2021 to reach the line — which is the month in which the Sahm and Michez indicators both peak. The instrument the San Francisco Fed dismisses agrees with the other

two on timing to the month. The 70-percent labor-force figure is a fair objection on breadth and is recorded as such.

“Several instruments never fired at all.” True, and they should be named. The Chicago Fed National Activity Index’s three-month average — the series its published rule reads, the Bank stating that “periods of economic contraction have historically been associated with values of the CFNAI-MA3 below -0.70 ” — never fell past -0.39 . The Chauvet–Piger smoothed recession probability, published on FRED and re-estimated with each release, never exceeded 2.3 percent on the August 2026 vintage used here. Hamilton’s GDP-based index, which calls a recession when “the value of the index rises above 67%,” peaked at 37.4. The Conference Board’s leading index stopped signaling recession in 2024 — its release of August 19, 2024 stated that “For the fourth consecutive month, the US LEI has not signaled a recession ahead” — after having forecast one for Q3 2023 through Q1 2024. And the SOS touched but never exceeded its own line. Three of these five are output- or activity-based composites, and the episode’s distinguishing feature is that it was concentrated in labor demand and rolled across sectors rather than striking them together — the configuration most likely to leave an activity composite unmoved. Ash and Nickelsburg (2024), in *Economics Letters*, make the general point in the Sahm rule’s own case. Analyzing it through historical episodes and stochastic simulations in a VAR framework, they conclude that “the Sahm rule is a poor predictor of future recessions,” that this follows from “its focus on the unemployment rate, which misses non-labour market shocks,” and that “the Sahm rule should be restricted to the purpose that its creator intended.” The lesson generalizes: an instrument keyed to one channel is silent about the channels it does not read, and silence from an activity composite is not evidence that a labor-demand contraction did not occur.

“Payroll employment never fell, and real GDP grew.” Both are true and neither is decisive. The NBER’s own criteria treat depth, diffusion, and duration as “somewhat interchangeable,” and the benchmark revisions removed 598,000 and 862,000 jobs from the March 2024 and March 2025 levels after the fact. On output, the paper does not claim real GDP fell; it claims that real GDP *as published at the time* did, that real GDI fell in two quarters of 2022 and still does, and that two of the six monthly indicators the NBER names fell in the first half of 2022 and still do.

“The NBER’s own six monthly indicators rose during the months you date.” They did, and this is the strongest single objection to the dating proposed here. Between April and August 2024 real manufacturing and trade sales rose 1.87 percent, real personal consumption expenditures 1.26 percent, real personal income less transfers 0.78 percent, industrial production 0.19 percent, and nonfarm payroll employment 0.14 percent; only household-survey employment fell, by 0.01 percent, a decline of nine thousand jobs. The contrast with dated recessions is stark. In the first four months of the 1990–91 recession, and again in the first four months of the 2001 recession, every one of the five for which monthly data reach that far back fell; in the first four months of the

2007–09 recession, five of the six fell. On the committee’s own instrument panel, April to August 2024 does not look like the opening of a contraction.

Three considerations qualify that without dissolving it. The two series that grew at rates resembling an expansion — sales at 5.7 percent and consumption at 3.8 percent at annual rates — are product-side measures, and it is the labor side that this paper argues contracted: the two employment measures were flat to falling, and industrial production, which had peaked in April 2022 and would not regain that level until mid-2025, gained six-tenths of a percent at an annual rate. The payroll gain across the four months was 227,000, smaller than the 598,000 that the annual benchmark later removed from the March 2024 level, so the direction of that series over this window is within the magnitude of its own subsequent revision. And the diffusion criterion cuts in both directions: a contraction that rolls through sectors while others expand — the shape Section 8 describes, and the one Brave and co-authors document on the labor-demand side — is precisely the configuration that leaves an activity panel unmoved while the labor market deteriorates beneath it.

“Menzie Chinn has already refuted the 2022 recession claim.” He refuted a different claim. Antoni and St. Onge (2024) argued from an alternative deflator that the United States had been in recession continuously since 2022; Chinn’s critique — the deflator is not reproducible, the housing treatment departs from theory, the broad-based criterion is not met — lands on that claim, which this paper does not make. Nothing here rests on a non-standard deflator or asserts a four-year recession. What Chinn concedes, and what this paper uses, is his own sentence: “manufacturing and trade sales as well as personal income ex.-current transfers dropped in 2022H1.”

10. Conclusion

The NBER-defined recession is not the only kind of recession, and the post-2022 cycle is the best evidence yet. A central bank that dismissed a forty-year-high inflation as transitory, kept buying bonds into the spring of 2022, and then delivered the steepest tightening since 1981 produced exactly what the textbooks predict: the longest yield-curve inversion on record, half a year of contracting real-time output, declines in two of the six monthly indicators the arbiter’s own procedure names that survive revision, a 39 percent collapse in job openings, a Sahm Rule crossing on both the data of the day and the data of today, a published vacancy-gated refinement of that rule at or above its recession line for fifteen consecutive months from March 2024, a thirty-state breadth rule at or across a line its authors say has never been crossed outside a recession, and a layoff-based indicator resting exactly on its trigger line. Screening seventy-seven years of data, crossings like 2024’s essentially do not happen outside dated recessions — and on the real-time record, 2024 is the only sustained one that ever has.

The parsimonious reading is that something recessionary happened — policy-induced, rolling, shallow, oddly shaped, and partially erased in revision — and the chart should say so. Hence the dotted gray bar, and it is a 2024 bar, not a 2022–24 one: onset April 2024 by the Sahm Rule — March 2024 on the Michez rule’s own dating — end August 2024 by the Bristow Rule — four months, briefer than every postwar contraction except the pandemic’s and the mildest of them all — drawn in dots rather than ink because the call belongs, in the end, to the committee that declined to make it. The pairing is the paper’s methodological point: the Sahm Rule dates a recession’s beginning, the Bristow Rule dates its end, and together they bracket the business cycle mechanically, in close to real time, from a single monthly data series — no committee, no revisions, no waiting.

The benchmark for what a well-run macroeconomy looks like is not hard to find: in 2017–2019 the unemployment rate fell to 3.5 percent, its lowest level since December 1969, with consumer price inflation averaging about 2.1 percent — an expansion ended by a pandemic, not by policy. Against that standard, the record of 2021–2025 is a policy failure twice over, first of diagnosis and then of dosage, redeemed by cushions the Federal Reserve did not build. For analysts and decision-makers the practical lesson is older than this episode: decisions are made on real-time data, recessions are dated on revised data, and anyone whose livelihood depends on the difference should watch a broad, fast, and varied set of indicators rather than wait for the gray bar.

Acknowledgments

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AI disclosure: Claude (Opus 5 and, for the September 2026 verification audit, Fable 5.1; Anthropic) was used in preparing this paper: drafting and revising manuscript text, writing the data-assembly and estimation code, and producing the figures from the deposited data. All research questions, specifications, thresholds, and pre-registrations were set by the authors. No figure was generated by a model from a prompt; each is plotted by deposited code from the deposited data, by the reproducible methods reported in Appendix A.3. All series were retrieved directly from FRED and the primary publishers in August 2026, and every statistic and figure in the paper was recomputed from the primary data. The authors reviewed and edited all text, verified every quotation and cited source against the primary document, and take full responsibility for the content, including any errors. A section-by-section account of AI use, with the model and version used at each stage, appears in Appendix A.4. Interpretations, views, and any remaining errors are the authors’.

Appendix A. Data, Reproducible Methods, and AI Use

This appendix is the one referred to in the AI disclosure above. It records every series used, the exact form of every rule, the reproducible method by which each figure is drawn, and a section-by-section account of AI use naming the model, version, and developer at each stage.

A.1 Series inventory. Every series was retrieved as a CSV from the Federal Reserve Bank of St. Louis in August 2026, from the FRED graph endpoint, which returns the full published history of the named series (fredgraph.csv, keyed on the series identifier in the first column of Table A1). No series was edited, spliced, or reconstructed. The only interpolation performed anywhere in the paper is the single missing October 2025 unemployment observation, described in Section 2 and in A.2 below.

Table A1. Series used, with source and span as retrieved

FRED ID	Series	Frequency	Span retrieved
UNRATE	Civilian unemployment rate	Monthly	Jan 1948 – Jul 2026
T10Y2Y	10-year minus 2-year constant-maturity Treasury spread	Daily	Jun 1976 – Aug 2026
GS10, GS1	10-year and 1-year constant-maturity yields	Monthly	Apr 1953 – Jul 2026
GDPC1	Real gross domestic product	Quarterly	1947 Q1 – 2026 Q2
A261RX1Q020SBEA	Real gross domestic income	Quarterly	1947 Q1 – 2026 Q2
W875RX1	Real personal income excluding current transfer receipts	Monthly	Jan 1959 – Jul 2026
CMRMTSPL	Real manufacturing and trade industries sales	Monthly	Jan 1967 – Jun 2026
INDPRO	Industrial production, total index	Monthly	Jan 1919 – Jul 2026
PAYEMS	All employees, total nonfarm	Monthly	Jan 1939 – Jul 2026
JTSJOL	Job openings, total nonfarm	Monthly	Dec 2000 – Jun 2026
CLF16OV	Civilian labor force level	Monthly	Jan 1948 – Jul 2026
IURSA	Insured unemployment rate, seasonally adjusted	Weekly	Jan 1971 – Aug 2026
FEDFUNDS	Effective federal funds rate	Monthly	Jul 1954 – Jul 2026
CPIAUCNS	Consumer price index, all urban consumers, not seasonally adjusted	Monthly	Jan 1913 – Jul 2026
WALCL	Total assets of the Federal Reserve	Weekly	Dec 2002 – Aug 2026
SAHMREALTIME	Sahm rule recession indicator, real-time	Monthly	Dec 1959 – Jul 2026
SAHMCURRENT	Sahm rule recession indicator, current vintage	Monthly	Mar 1949 – Jul 2026
USREC	NBER-based recession indicator	Monthly	Dec 1854 – Jul 2026

Replication materials. References in this appendix to deposited code and deposited data are to the archive accompanying this paper, whose contents, provenance and reproduction instructions are set out in Appendix B.

Notes: Eight further FRED series enter individual calculations rather than the rules themselves: the target-range bounds (DFEDTARU, DFEDTARL) in Section 3, real personal consumption expenditures (PCEC96) and household employment (CE16OV) in Section 9, and the four instruments discussed there (CFNAI, CFNAIMA3, RECPROUSM156N, JHGDPBRINDX). All eight are named in the FRED reference entry and deposited in the archive. Non-FRED primary sources are cited at the point of use in the text and enumerated in the references: the Bureau of Economic Analysis advance estimates of April 28 and July 28, 2022; the Bureau of Labor Statistics Employment Situation of August 2, 2024 and the CES benchmark articles of February 7, 2025 and February 11, 2026; the NBER business-cycle chronology and dating-procedure FAQ; the Richmond Fed SOS indicator page; the Michailat and Saez Automated Business Cycle Dashboard, whose `recession_indicator.csv` and `recession_probability.csv` files were retrieved in August 2026; the San Francisco Fed Economic Letter 2025-19 and Working Paper 2025-31; and the Conference Board, NAHB, Cass and ISM releases named in the text.

A.2 Rules, in the exact form computed. Let $u(t)$ be the monthly unemployment rate and $\bar{u}(t)$ its three-month moving average.

Sahm Rule. The indicator is $S(t) = \bar{u}(t) - \min\{\bar{u}(t-12), \dots, \bar{u}(t-1)\}$ — the current three-month average less the lowest three-month average of the twelve months preceding it. A recession is signaled when $S(t) \geq 0.50$. This convention, and not one that includes the current month in the minimum, is the one FRED publishes: recomputed from UNRATE with the October 2025 interpolation described below, it reproduces SAHMCURRENT exactly in 928 of the 929 months on record; the single exception is October 2025 itself, for which FRED’s published value averages the available months of the three-month window rather than an interpolated one. Both the real-time series (SAHMREALTIME) and the current-vintage series (SAHMCURRENT) are used as published by FRED; neither was recomputed for the screen, so the screen inherits FRED’s own construction. Where the October 2025 unemployment observation is required by a rolling window, the missing value is replaced by linear interpolation between September and November 2025; it enters no other calculation and no reported statistic depends on it.

Bristow Rule. A crossing episode is a maximal run of consecutive months with $S(t) \geq 0.50$. The end date is $\arg \max S(t)$ over the window running from the episode’s first month to twelve months after its last, ties resolved to the first occurrence. Where more than one such run begins inside a recession’s window — as the real-time series does in 1974, crossing for a single month in March before the sustained run from July — the episode is the longest. The window is defined by the indicator alone, so the rule requires no official dates and can be applied to an episode the committee has not dated; run on the twelve postwar recessions it returns exactly the same twelve dates as a window anchored on the NBER peak and trough, and the record reported in Section 2 is unchanged under either. The confirming second-derivative statistic is $\arg \max [S(t) - S(t-3)]$ over the same window. The onset lag reported in Section 2 is the number of months from each NBER

peak to the first month in which $S(t) \geq 0.50$, computed over the window from the peak to six months after the trough.

Michez rule. With $v(t)$ the vacancy rate — job openings (JTSJOL) divided by the civilian labor force (CLF16OV), in percent — and $\bar{v}(t)$ its three-month moving average, the two channels are $\hat{u}(t) = \bar{u}(t) - \min\{\bar{u}(t-12), \dots, \bar{u}(t-1)\}$, the Sahm indicator of the paragraph above, and $\hat{v}(t) = \max\{\bar{v}(t-12), \dots, \bar{v}(t-1)\} - \bar{v}(t)$, and the indicator is $m(t) = \min\{\hat{u}(t), \hat{v}(t)\}$. Recession is signaled at $m > 0.29$ percentage point, certain above 0.81, with probability $(m - 0.29)/(0.81 - 0.29)$ in between. The readings quoted in Section 7 and plotted in Figure 4 are the authors' own published dashboard series. The independent recomputation described there applies the formula above to JTSJOL, CLF16OV and UNRATE; because the dashboard assigns each month the previous month's job openings and computes unemployment at basis-point precision, that recomputation reproduces the shape and the peak month of their indicator but not its exact values; with those two conventions applied it reproduces the dashboard series, and the unshifted recomputation is reported separately so the two are never conflated.

SOS indicator. With $c(t)$ the weekly insured unemployment rate (IURSA) and $\bar{c}(t)$ its 26-week moving average, the indicator is $\bar{c}(t) - \min\{\bar{c}(t-52), \dots, \bar{c}(t-1)\}$, the minimum being taken over the 52 weeks preceding the current one. A recession is signaled at a rise of more than 0.20 percentage point. This construction reproduces the Federal Reserve Bank of Richmond's own published SOS series exactly, in all 2,826 weekly observations from June 1972 to August 2026; the readings quoted in Section 6 and plotted in Figure 3 are that published series. The twelve-week plateau reported there is the set of weeks for which the indicator equals 0.2000 to four decimal places.

Yield-curve statistics. An inversion run is a maximal set of consecutive published observations of T10Y2Y with a value strictly below zero; its length is counted in observations, which are trading days. The cumulative area reported in Section 3 is the sum of the daily values over a run, in percentage-point days.

A.3 How each figure is drawn. No figure was generated by a model from a prompt. Each is plotted by deposited Python code — written with Claude Opus 5, developed by Anthropic, and reviewed by the authors — from the deposited CSVs, using pandas 3.0.2 for the transformations and matplotlib 3.10.9 for the drawing, at 200 dots per inch, with NBER recession shading taken from the chronology in Table A1's USREC entry and the published NBER peak and trough dates.

Figure 1 plots T10Y2Y as published, unsmoothed, with a zero line, over June 1976 to August 2026. The annotated run length, trough, and comparison inversions are the quantities defined in A.2, computed in the same script that draws the panel.

Figure 2 plots UNRATE in the upper panel and SAHMCURRENT in the lower, 1948 to July 2026, with the +0.50 threshold drawn as a dashed line. The dotted boxes are drawn at the dates the two rules produce, not placed by hand. The marked point in the lower panel is the August 2024 value.

Figure 3 plots the SOS indicator, computed from IURSA by the formula in A.2 and verified against the Richmond Fed’s published series, with the +0.20 threshold drawn as a dashed line, 1972 to August 2026.

Figure 4 plots the Michailat and Saez dashboard recession_indicator.csv against SAHMCURRENT from June 2021, with the axis extended to September 2026 and the data ending in July 2026, each with its own threshold line.

Recession shading in every figure marks the twelve NBER contractions from November 1948 to April 2020 that fall inside the plotted window.

A.4 Section-by-section account of AI use. The model used at every stage below was Claude Opus 5, developed by Anthropic; the independent verification audit of September 2026, which produced the corrections carried in this version, was run on Claude Fable 5.1, also developed by Anthropic. No other model was used. All research questions, specifications, thresholds, and dating conventions were set by the authors. Table A2 records the division of labor stage by stage.

Table A2. AI use by stage

Stage	What the model did	What the authors did
Data assembly	Wrote and ran the download scripts for the nineteen FRED series in Table A1 and for the dashboard CSVs	Specified which series, at what frequency, and on which vintage basis
Figures	Wrote the matplotlib code that draws Figures 1 to 4 from the deposited CSVs	Specified what each figure shows and reviewed the rendered output
Drafting	Drafted and revised manuscript text against the verified material, preserving the structure and voice of the earlier drafts	Set the argument, the claims, and the conclusions; reviewed and edited all text

No model output is an analyzed input to any result in this paper: no hypothesis was generated by a model, and no variable that enters a computation was created by one. Under the Journal of Labor Economics rule, which is the strictest of those surveyed, that is the condition under which an in-text citation and reference entry for the model would become required; a reference entry is nonetheless provided. Every quotation, figure, date and threshold in the paper was checked against the primary document or recomputed from the deposited series before publication, and a specific that could not be so verified was deleted rather than qualified.

Appendix B. The Replication Archive

The archive accompanying this paper is a single compressed folder. It is the complete set of materials required to regenerate every number and every figure reported here without contacting the authors. Its permanent identifier is to be inserted at publication.

B.1 Contents. | Path | What it holds |

<code>`data/fred/`</code>	Twenty-seven CSV files, one per FRED series, exactly as retrieved in August 2026
<code>data/dashboard/</code>	The Michailat and Saez dashboard indicator and the Richmond Fed SOS series, as published by their authors
<code>data/robustness/</code>	The pre-1948 unemployment series used for the 1929–33 and 1937–38 tests in Section 2 (M0892AUSM156SNBR), together with OECD unemployment and recession-indicator series for Canada, Germany, Japan and the United Kingdom that were retrieved for non-US tests not reported in this paper
<code>code/reproduce.py</code>	Recomputes every statistic reported in this paper and prints each beside the value stated here
<code>code/figures_1_2.py</code> , <code>code/figures_3_4.py</code>	Draw Figures 1 to 4 from the deposited CSVs
<code>code/build_manuscript.py</code>	The typesetting script that produces the manuscript file
<code>figures/</code>	The four figures as they appear above
<code>manuscript/</code>	The manuscript source and the paper as circulated
<code>README.md</code> , <code>README.docx</code>	Provenance, the exact form of every rule, and the limitations stated in Section 2, in plain text and in Word format
<code>MANIFEST.txt</code>	A SHA-256 hash for every file in the archive

B.2 Reproducing the results. From the code directory, `python3 reproduce.py` recomputes 110 reported statistics from the deposited data and prints each against the value stated in this paper, exiting zero only if all of them reproduce. It requires Python 3.11 or later with `pandas` and `matplotlib`, and runs in a few seconds. On the deposited data all 110 reproduce.

The figures are redrawn by the two figure scripts, which regenerate the deposited PNG files byte for byte, and `build_manuscript.py` rebuilds the circulated file from the deposited source and those figures. Every path in every script resolves relative to the script’s own location, so the archive reads nothing outside itself.

B.3 Provenance. Every FRED series was retrieved from <https://fred.stlouisfed.org/graph/fredgraph.csv?id=<SERIES>>, which returns the full published history of the named series. No series was edited, spliced, or reconstructed; the files are byte-for-byte as downloaded. The download was performed four times on separate occasions during verification, and the four copies were byte-identical. The single interpolation described in Section

2 — the unpublished October 2025 unemployment observation — is applied in code, not in the deposited data, so the CSVs remain exactly as published.

Every quotation in this paper was verified against the primary document it cites. Those documents are not redistributed in the archive, for copyright reasons; each is identified in the reference list.

References

- Anthropic. 2026. *Claude* (Opus 5; Fable 5.1 for the September 2026 verification audit) [large language model]. <https://claude.ai>. Used for research assistance, data compilation, figure generation, and drafting; all outputs verified against the primary sources below.
- Antoni, E. J., and Peter St. Onge. 2024. “Recession Since 2022: US Economic Income and Output Have Fallen Overall for Four Years.” *Brownstone Journal*, October 9, 2024.
- Ash, Thomas, and Jerry Nickelsburg. 2024. “Works like a Sahm: Recession indicators and the Sahm rule.” *Economics Letters* 242, 111878. Quotations from the published abstract.
- Atkinson, Tyler. 2024. “Rising unemployment does not mean recession is inevitable.” *Dallas Fed Economics*, October 8, 2024. Federal Reserve Bank of Dallas.
- Blanchard, Olivier, Alex Domash, and Lawrence H. Summers. 2022. “Bad News for the Fed from the Beveridge Space.” Peterson Institute for International Economics, Policy Brief 22-7, July 2022.
- Blinder, Alan S. 2023. “Landings, Soft and Hard: The Federal Reserve, 1965–2022.” *Journal of Economic Perspectives* 37 (1): 101–120.
- Brave, Scott, Erin Crust, Stefano Eusepi, Bart Hobijn, and Ayşegül Şahin. 2026. “Making Sense of Labor Market Indicators amid Data Imperfections.” *Brookings Papers on Economic Activity*, Spring 2026 (quotations from the conference draft of March 2026).
- Bureau of Economic Analysis. 2022. “Gross Domestic Product, First Quarter 2022 (Advance Estimate).” April 28, 2022; and “Gross Domestic Product, Second Quarter 2022 (Advance Estimate).” July 28, 2022. U.S. Department of Commerce.
- Bureau of Labor Statistics. 1960. *Analysis of Work Stoppages, 1959*. Bulletin 1278. Washington, DC: U.S. Department of Labor.
- Bureau of Labor Statistics. 2024. *The Employment Situation — July 2024*. Released August 2, 2024. U.S. Department of Labor.
- Bureau of Labor Statistics. 2025. “BLS Establishment Survey National Estimates Revised to Incorporate March 2024 Benchmarks.” CES National Benchmark Article, February 7, 2025. U.S. Department of Labor.
- Bureau of Labor Statistics. 2026. “BLS Establishment Survey National Estimates Revised to Incorporate March 2025 Benchmarks.” CES National Benchmark Article, February 11, 2026. U.S. Department of Labor.

- Burger, Albert E. 1969. “A Historical Analysis of the Credit Crunch of 1966.” *Review*, Federal Reserve Bank of St. Louis 51 (September): 13–30.
- Cass Information Systems. 2024. *Cass Transportation Index Report*, February 2024.
- Chinn, Menzie. 2024. “America In Recession since 2022? A Critique of Antoni-St. Onge.” University of Wisconsin–Madison, November 6, 2024. <https://users.ssc.wisc.edu/~mchinn/OnASO.pdf>
- Conference Board, The. 2023. “LEI for the U.S. Fell Further in June.” Press release, July 20, 2023.
- Conference Board, The. 2024. “The Conference Board Leading Economic Index® (LEI) for the U.S. Declined in July.” Press release, August 19, 2024.
- Demianczuk, John, Abhi Gupta, and Ryan Nunn. 2026. “What We Learned About Recession Indicators, Rules, and Triggers After the Pandemic.” The Budget Lab at Yale, August 10, 2026.
- de Soyres, François, Johannes Fleck, William Peterman, Avani Pradhan, and David Ratner. 2026. “Population Growth and Labor Market Fragility: Lessons from Domestic and International Experiences.” FEDS Notes, Board of Governors of the Federal Reserve System, May 20, 2026.
- Federal Reserve Bank of Chicago. “Chicago Fed National Activity Index: About the CFNAI.” Accessed September 2026.
- Federal Reserve Bank of Richmond. 2026. “Scavette–O’Trakoun–Sahm-style (SOS) Recession Indicator.” Accessed August 2026.
- Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis. Series UNRATE, T10Y2Y, GS10, GS1, GDPC1, A261RX1Q020SBEA, W875RX1, CMRMTSPL, INDPRO, PAYEMS, JTSJOL, CLF16OV, IURSA, FEDFUNDS, CPIAUCNS, WALCL, DFEDTARU, DFEDTARL, SAHMREALTIME, SAHMCURRENT, USREC, PCEC96, CE16OV, CFNAI, CFNAIMA3, RECPROUSM156N, and JHGDGPBRINDX. Accessed August 2026.
- Garimella, Rohit, Òscar Jordà, and Sanjay R. Singh. 2025. “Tracking Labor Market Stress.” *FRBSF Economic Letter* 2025-19, August 18, 2025; and “A New Labor Market Stress Indicator.” Federal Reserve Bank of San Francisco Working Paper 2025-31, December 2025. Monthly and weekly series read from the Bank’s published Labor Market Stress Indicator data file, August 2026.
- Hornstein, Andreas. 2023. “Unemployment Changes as Recession Indicators.” *Richmond Fed Economic Brief* 23-13, April 2023.
- Institute for Supply Management. 2022. “Manufacturing PMI at 49%; November 2022 Manufacturing ISM Report On Business.” December 1, 2022; and “Manufacturing PMI at 48.4%; November 2024 Manufacturing ISM Report On Business.” December 2, 2024.
- Macaluso, Claudia. 2026. “Business Cycles: Is the Current Economic Environment Different?” *Richmond Fed Economic Brief* 26-07, February 2026.
- Michaillat, Pascal. 2025. “Early and Accurate Recession Detection Using Classifiers on the Anticipation-Precision Frontier.” NBER Working Paper 34015, July 2025; published as “Recession Detection Using Classifiers on the Anticipation-Precision Frontier,” arXiv 2506.09664 v4, December 2025.
- Michaillat, Pascal, and Emmanuel Saez. 2025. “Has the Recession Started?” *Oxford Bulletin of Economics and Statistics* 87 (6): 1047–1058. Automated Business Cycle Dashboard, read August 2026.
- National Association of Home Builders. 2022. “Housing Recession Deepens.” NAHB blog, August 25, 2022, reproducing the *Eye on the Economy* newsletter of NAHB Chief Economist Robert Dietz.
- National Bureau of Economic Research. “Business Cycle Dating”; “US Business Cycle Expansions and Contractions”; “Business Cycle Dating Procedure: Frequently Asked Questions”; “Business Cycle Dating Committee Announcements.”

- O'Trakoun, John, and Adam Scavette. 2025. "A better Sahm rule? Introducing the SOS recession indicator." *Economics Letters* 247, 112175; and "SOS! Signaling Recessions Earlier," Federal Reserve Bank of Richmond *Economic Brief* 25-07, February 2025.
- Powell, Jerome H. 2021. Testimony before the Committee on Banking, Housing, and Urban Affairs, U.S. Senate, November 30, 2021.
- Sahm, Claudia. 2019. "Direct Stimulus Payments to Individuals." In *Recession Ready: Fiscal Policies to Stabilize the American Economy*, edited by Heather Boushey, Ryan Nunn, and Jay Shambaugh. Washington, DC: The Hamilton Project and Washington Center for Equitable Growth. Real-time indicator: FRED series SAHMREALTIME.
- Sahm, Claudia. 2024. "Sahm-thing more on the Sahm rule." *Stay-At-Home Macro*, July 26, 2024; and "My recession rule was meant to be broken." *Bloomberg Opinion*, August 7, 2024.